Module 8 Instructional Techniques



Module objectives

At the end of this module you should be able to:

Describe	the sections of a lesson plan
Develop	an appropriate rationale for learning
Write	well-constructed learning objectives
Describe	interactive instructional strategies
Determine	appropriate instructional aids
Prepare	lesson plan





Lesson Plan

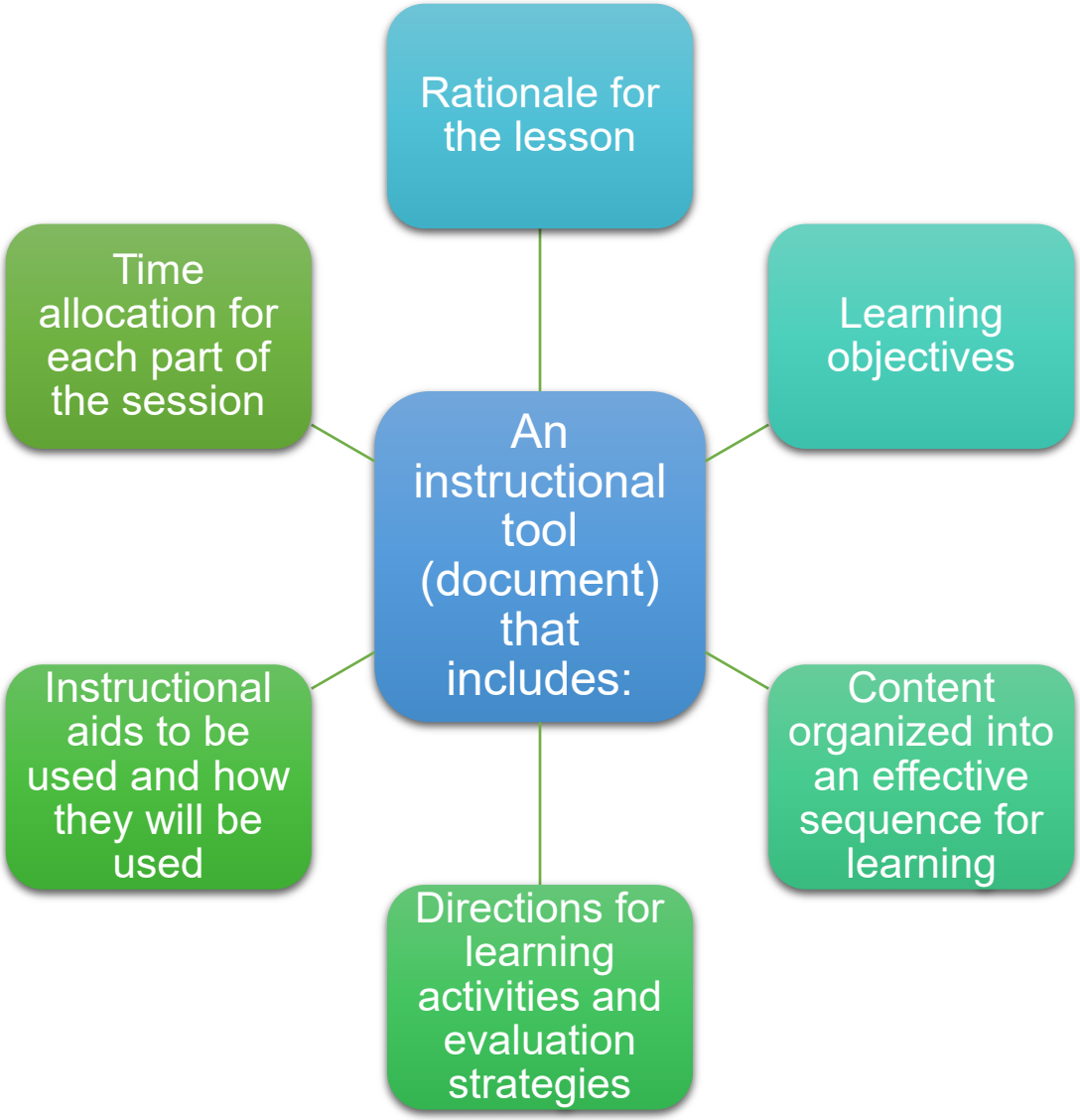
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Lesson plan

- A lesson plan is an instructor's detailed description of the course of instruction for one class
  - Details will vary depending on the preference of the instructor, subject being covered, and the need and/or curiosity of students

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A lesson plan IS



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Anatomy of a lesson plan

Lesson information	• Title, Timing, Rationale, Learning Objectives, Evaluation Strategy, Instructional Strategies, Instructional Aids
Start	• WIIFM • List of learning objectives
Conduct	• Description of lesson/course content, activities and exercises, and a list of instructional aids • Breakdown of time for each module and activity/exercise
Wrap-up	• Review of key points • Evaluation (if objective has been met) • Link this lesson with upcoming lesson, if relevant

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Lesson plan

- Our goal is to create an instruction that is:
  - Effective
  - Efficient
  - Engaging
- There should be a strong link among the instructional objectives, the learning activities, and learner assessment
- Learners should be evaluated in terms of how well they achieve the instructional objectives, instead of how they compare against fellow learners

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Developing a lesson plan

1 Start with the end in mind

- **Identify your** learning objective
- **Create your** lesson assessment
- What do you need to see from students to know if they mastered the objective?
- **Write your** key points
- Break down your objective into the bare essentials

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Developing a lesson plan

2 Choose your instructional methods

- Select your instructional methods
- Lecture, discussion group, simulation, case study, group project, etc.

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Developing a lesson plan

3 Adjust your plan

- Check the pacing and alignment of all parts of the lesson
- Can the lesson as a whole be completed in the overall amount of time?
- Is a realistic amount of time allocated to each part of the lesson?
- Is the lesson paced so that it devotes sufficient time to activities where students develop and demonstrate mastery

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5 Common mistakes

1. The Learning objective doesn't specify what the student will actually do that can be observed
2. The lesson assessment is disconnected from the behavior indicated in the objective
3. The materials specified in the lesson are extraneous to the actual described learning activities. Overkilling with materials is not a virtue!
4. The instruction methods in which the instructor will engage is not efficient for the level of intended student learning
5. The student activities described in the lesson plan do not contribute in a direct and effective way to the learning objective

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Course delivery assignment (day 3)

Exercise

- Teach a 10 minute lesson
- Demonstrate core instructor skills
- Take into account principles of adult learning and active training:
- Be prepared to provide:
  - Instructor scripts
  - Presentations to participants
  - Handouts
  - Exercises
  - Detailed outline & lesson plan

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Select a training topic

Exercise

- Select a training topic for which you will be developing a 10 m lesson for the final presentation
- Criteria for your training topic:
  - The topic is HF related
  - You are an expert on the content /skill

- Time: 5 minutes

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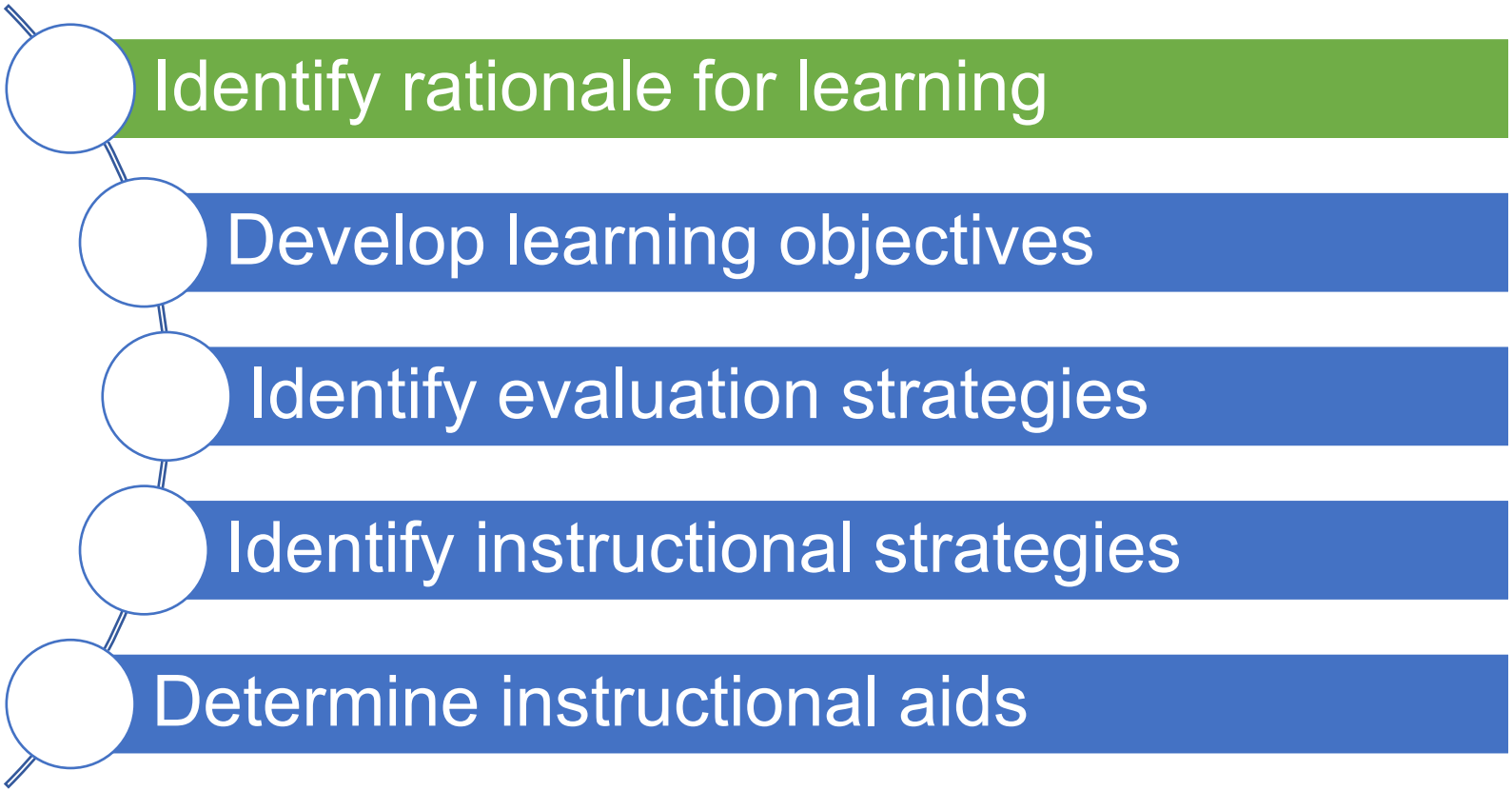
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Develop lesson plans



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Which is better?

- This workshop will enable you to:
  - Name your country's 10 wealthiest people
  - Determine five ways to double your income

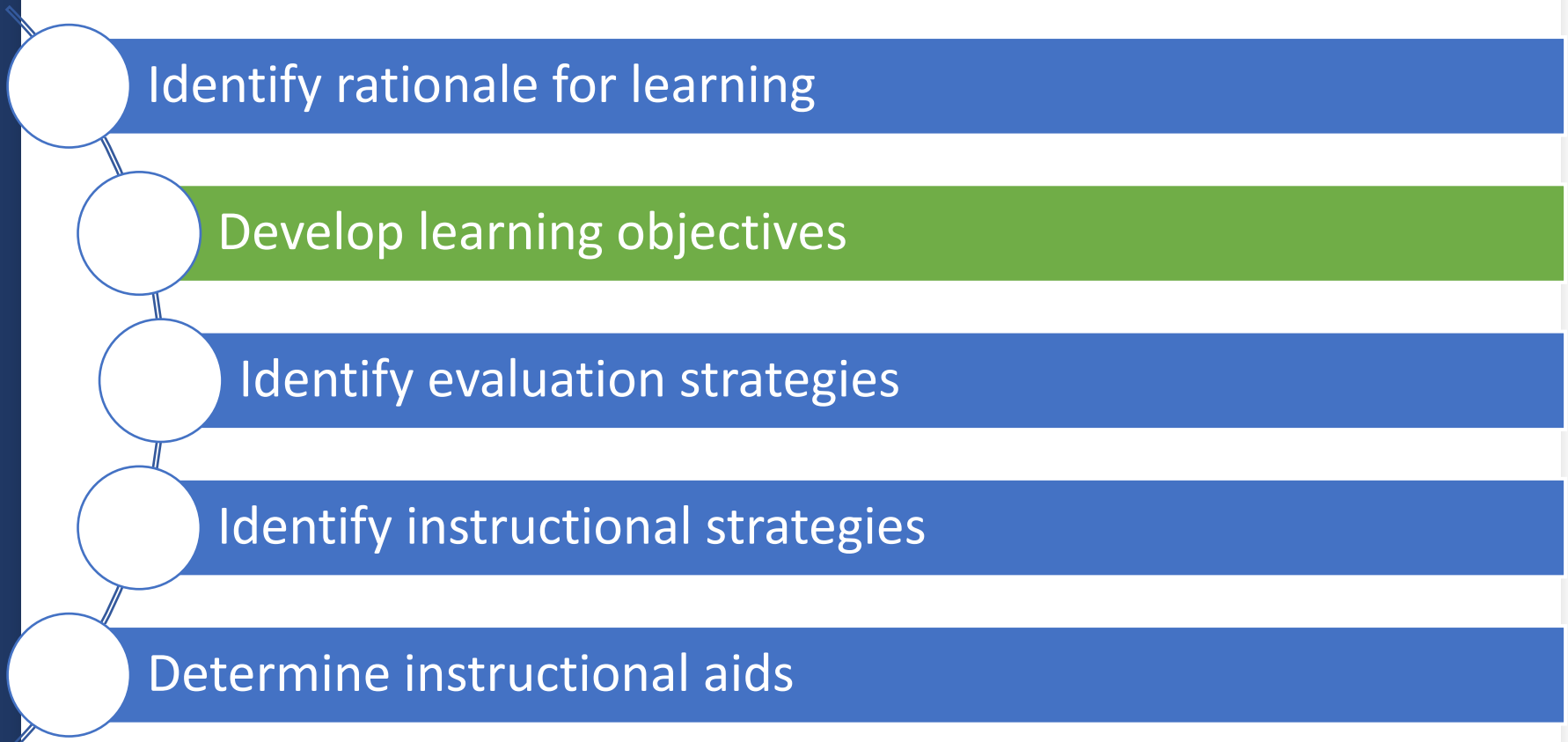
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Rationale for learning

- Adult learners need to be ready to learn. For this to happen, the instructor must help the learner answer the question, “What’s in it for me?” (WIIFM)

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Develop lesson plans



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What is a learning objective?

- Answer the question:
  - What is it that your participants should be able to do at the end of the class?
- Makes clear the intended learning outcome
- Focus on student performance

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Why learning objectives?

Serves to connect content and assessment around learning

Guides selection of teaching/learning activities that will best achieve objectives

Gives learners a clear picture of what to expect and what's expected of them

Forms the basis for evaluating teacher, learner, and curriculum effectiveness

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Key component of learning objectives

SPECIFIC
knowledge, skills and attitudes of a learner

ACTION ORIENTED
Contain verb that demonstrate acquisition of knowledge

TIME-BOUND
A timeframe in which a learner is expected to achieve objective

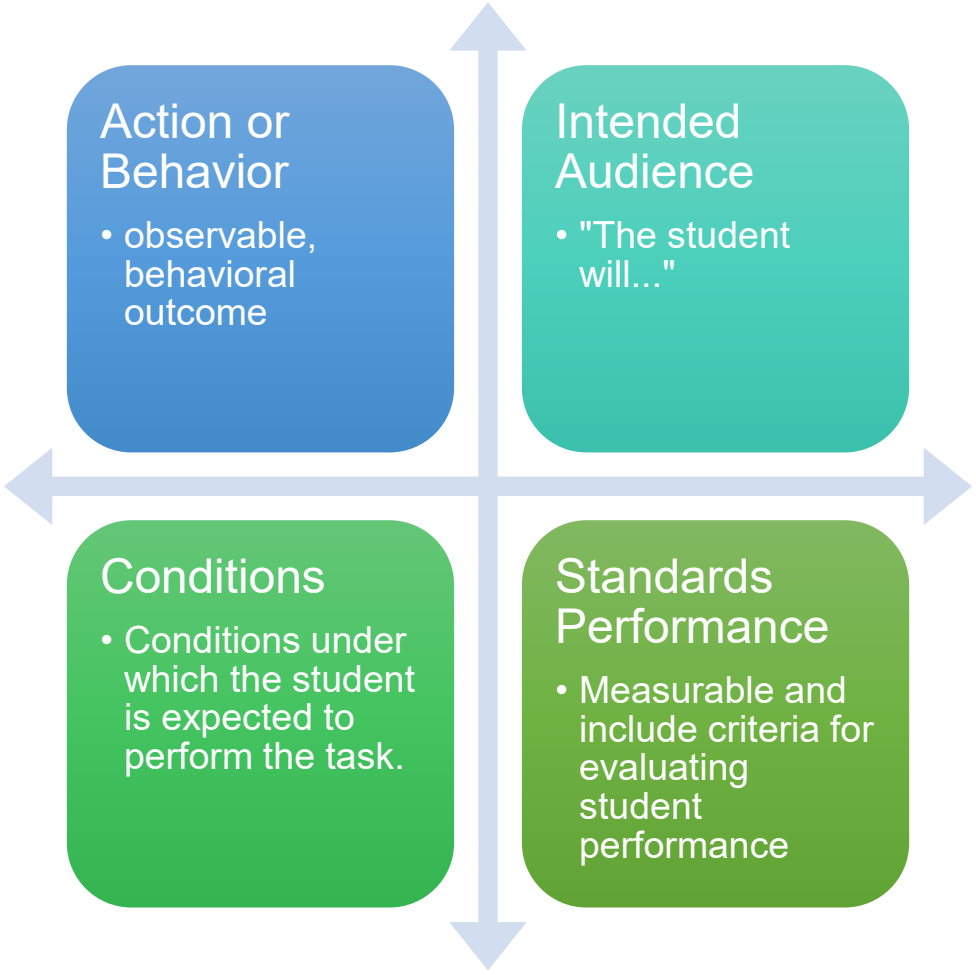


MEASURABLE
Achievement of learning can be measured

REASONABLE
Objective must be a reasonable and realistic expectation

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Guidelines for Writing effective Learning Objectives



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Example of good learning objectives

CONDITION

INTENDED AUDIENCE

Following this training, the participants will be able to complete every section of Dangerous Goods Acceptance checklist correctly

ACTION/ BEHAVIOR

STANDARD PERFORMANCE

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Action or behaviour

- Objectives need to reflect what a trainee will be able to do as a result of the training
- Objectives need to describe an action that a trainee will be able to perform
- An action is something that can be observed

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Which is better?

- The trainee will know how to accurately reconcile a bank statement in ten minutes
- The trainee will list the steps required to reconcile a bank statement accurately

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Words NOT to use

- Comprehend
- Grasp
- Know
- Learn
- Discern
- Understand



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Action words to use

- Write
- List
- Explain
- Demonstrate
- Operate
- Sketch
- Compare
- Construct



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Standards of performance

- Standards define how well you expect your trainees to perform the new skill or job at the end of the training
- Standards in your learning objective should match the on-the-job standards

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Standards - examples

- The trainee will be able to operate a computer program with no errors
- The trainee will be able to accurately reconcile a bank statement in ten minutes
- The trainee will list the steps for fixing an air conditioning unit on the first attempt

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Conditions

- Refer to what the learner will be given, and under what circumstances they will be asked to perform the action
- Include such things as the equipment available, a person present, a set of instructions, physical environment, location etc.

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Conditions - examples

- Given a reference manual, the trainee will be able to operate a computer program with no errors
- Given a calculator, the trainee will be able to accurately reconcile a bank statement in 10 minutes
- The trainee will list the steps for fixing an air conditioning unit without referring to the course materials on first attempt

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Identify “Good” Learning Objectives

Exercise

- Work in pairs
- See handout for further details

Time: 5 minutes

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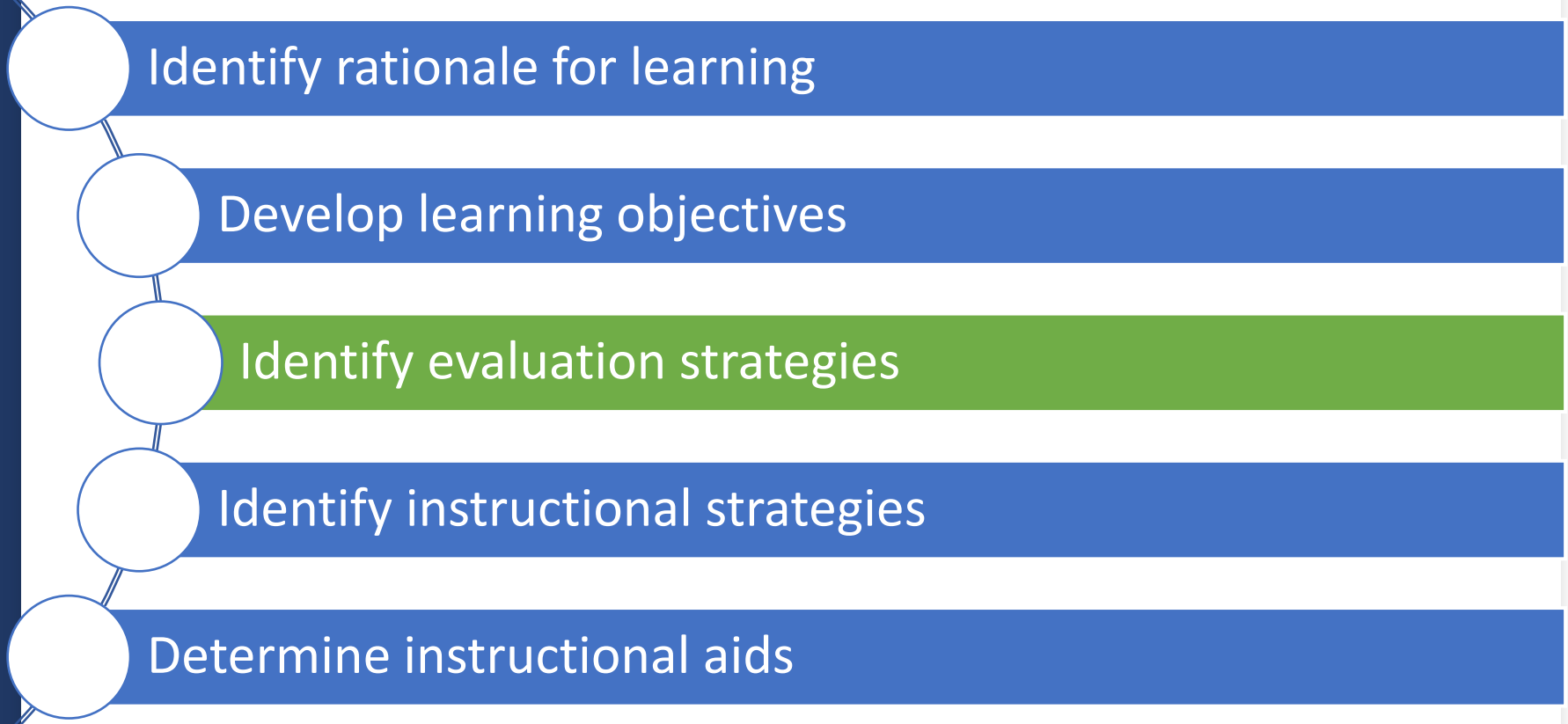
Write a learning objective for the topic that you selected for your final presentation

- Work individually
- See handout for further details

- Time: 10 minutes

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Develop lesson plans



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Determine evaluation strategy

- How are you going to measure whether or not trainees are able to do what is described in the objective?

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Determine evaluation strategy

Is it possible for a student to fail a CRM or HF training session?

How do you measure a students knowledge of the subject?

Are there right and wrong answers when facilitating CRM?

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General strategies

- Focus on learning outcomes
- Decide what to assess
- Aim for validity and reliability
  - Validity
  - Reliability
- Use variety of testing formats

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Evaluation measures

- Instead of testing, measure what the students know
- Ask open and leading questions about particular subjects
- Use open discussion techniques
- Pay attention to what is said, make notes of student performance during the session

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Evaluation measures

- If students are not able to demonstrate a sound understanding of a subject utilize input from others to reach a suitable answer
- Build student confidence
- True evaluation of CRM understanding will be in the workplace

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Develop lesson plans

- Identify rationale for learning
- Develop learning objectives
- Identify evaluation strategies
- Identify instructional strategies
- Determine instructional aids

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Instructional strategies

- Instructional strategies determine the approach a teacher may take to achieve learning objectives

- Instructional methods are used by teachers to create learning environments and to specify the nature of the activity in which the teacher and learner will be involved during the lesson

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Determine instructional strategy

- How to deliver content?
- How learners will learn the content?
- How you will give feedback?

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Methods of instruction



Direct Instruction
teaching by giving spoken explanations of the subject that is to be learned



Inductive, learner centered, and activity oriented
Experiential Learning



Interactive Instruction
Discussion and sharing among participants



Student learn independently by solving real-life problem
Indirect Instruction

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Actively involve learners

- Adult learners need to play an active role in training sessions; they need to do something more than listen or watch you during the session

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Active training

- What do you think makes training active?
- How does it compare to training that you have been involved in previously?
- How is active training learner-focused?

Time: 5 minutes

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Active training techniques

- Discussions
- Hands-on practice
- Pen and paper exercises
- Group assignments
- Case studies
- Role plays
- Triggers, jolts and games
- Simulations

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Using the appropriate instructional methods

To Convey Information
- Lecture
- Discussion Group
- Demonstration

To Convey difficult subject
- Discussion Group
- Simulation
- Debate
- Research assignments

To involve people
- Discussion
- Role playing
- Case studies
- Group work

TO POOL THOUGHTS AND IDEAS
- Discussion
- Brainstorming
- Group work

TO REINFORCE MEMORY
- Drill
- Memory aids
- Practice
- Written work

TO TEACH A SKILL
- Demonstration
- Simulation

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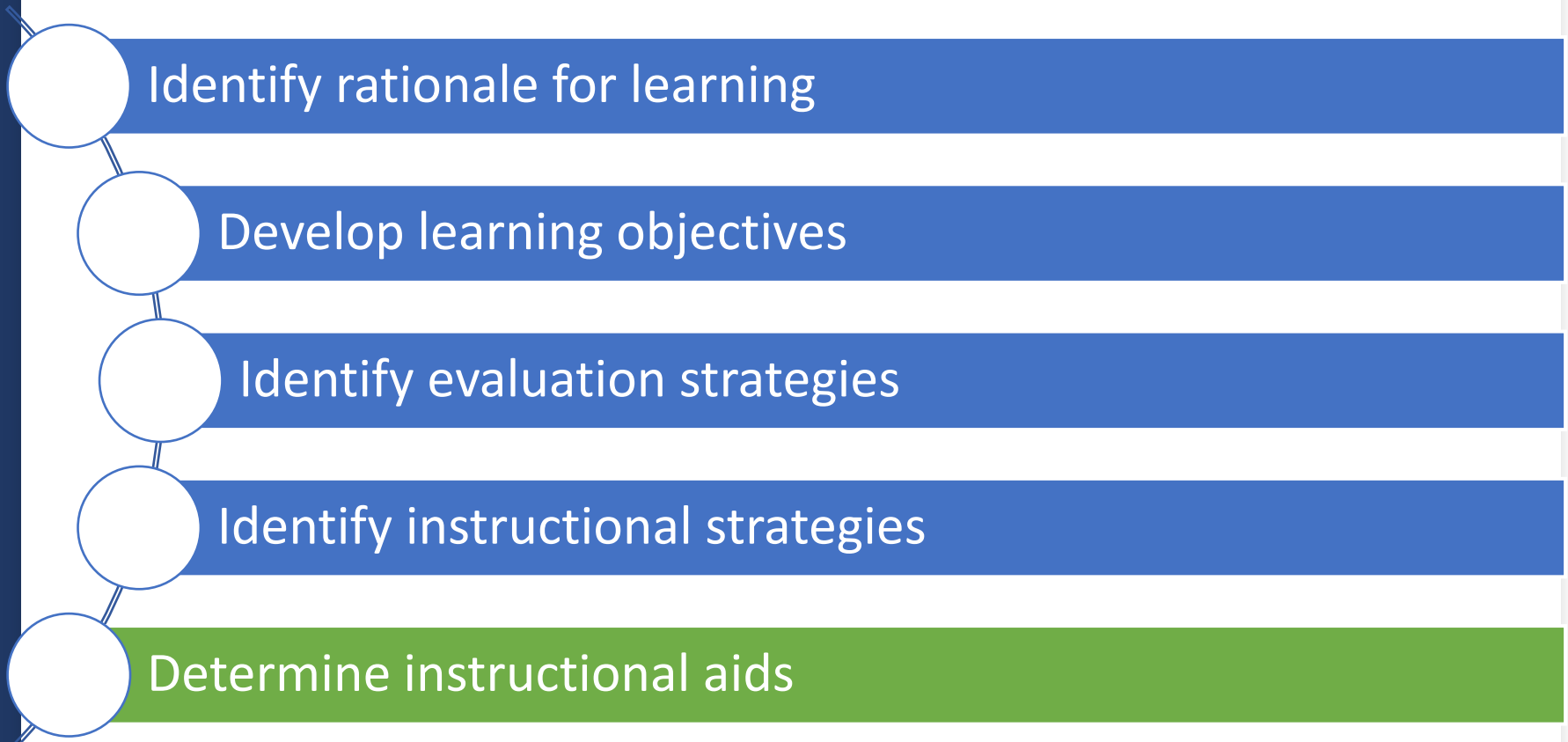
Your comfort level

- Are you comfortable playing games?
- Do you think they are appropriate for adults?
- What do you do you if you have reluctant participants?

Time: 5 minutes

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Develop lesson plans



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Use instrusctional aids

- Instructional aids are any device, object, equipment, machine, mock-up, illustration, simulator or other item that improves the teaching / learning process

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Use instructional aids

- Instructional aids can be used to:
  - Focus attention
  - Complement and reinforce verbal messages (not just repeat them)
  - Stimulate interest
  - Illustrate concepts, principles, etc., that are more difficult to internalize
  - Provide practice



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Use instructional aids

- Do Not use instructional aids to:
  - Impress audiences
  - Avoid interaction with participants
  - Replace preparation



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Develop a lesson to teach the other team how to build and fly a paper airplane

Exercise

- Work in groups
- See handout for further details

Time: 15 minutes

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Questions?



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